

## Sampling Strategies for Assessing Hydraulic Conductivity and Water-Conducting Voids in Saprolite

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### ABSTRACT

A saprolite's saturated hydraulic conductivity ( $K_{sat}$ ) and types of water-conducting voids must be known to assess its suitability for liquid waste disposal. This study determined optimum numbers of pits and numbers of samples per pit needed to estimate a saprolite's  $K_{sat}$  and volume percentages of water-conducting macrovoids and groundmass (microvoids). Three pits were dug across a 5-ha area to expose B horizons and saprolite (2-m depth), and undisturbed cores were collected from each pit. The  $K_{sat}$  was measured for each core, and water-conducting voids were dyed. Volume percentages of water-conducting (dyed) macrovoids (e.g., root channels with diam.  $\geq 0.1$  mm) and water-conducting groundmass were determined from thin sections. Mean values for saprolite  $K_{sat}$  and volume percentages of water-conducting macrovoids and groundmass were  $0.25 \text{ cm h}^{-1}$  and 1.1 and 47.9%, respectively. In order to minimize both the variance of each mean value and the time required for analysis, it was most efficient to collect only one core per pit. The most efficient number of pits depended on the desired size of the confidence interval and the property of interest. A 95% confidence interval for saprolite  $K_{sat}$  that was within 10% of the true mean would require over 700 pits per 5 ha, while an estimate of the percentage of water-conducting groundmass would require only 10 pits per 5 ha. Increasing the width of the 95% confidence intervals to be within 50% of the means reduced the required number of pits for these properties to 28 and 1, respectively.

SAPROLITE is a general term for weathered bedrock that has developed the friable consistence (when moist) of soil but which still retains some of its original rock structure (Pavich, 1986). Saprolite underlies many of the soils of the Piedmont region of the southeastern USA. Deposits may occur within 1 m of the soil surface and extend to depths exceeding 6 m (Daniels et al., 1984). Because saprolite is porous, it is being considered for use in the on-site disposal of household liquid wastes in Piedmont areas of North Carolina that are not served by municipal sewers and where soils are too thin for conventional septic systems (Amoozegar and Hoover, 1989).

The suitability of soil or saprolite for waste disposal is determined, in part, from the material's  $K_{sat}$ . Determining the  $K_{sat}$  for materials at individual homesites is impractical because the measurements are time consuming and expensive, and the number of homesites too great. An alternative approach is to determine mean  $K_{sat}$  values for specific areas of soil or saprolite, and then extrapolate these results to other similar sites using morphological characteristics such as texture, structure, large voids, or other features.

The precision with which mean values for  $K_{sat}$  and

morphological characteristics can be estimated depends on the numbers of samples collected and their locations. Identifying numbers and locations of samples needed to minimize the variance of the mean values can be done using analysis of variance procedures (Petersen and Calvin, 1986). The variance of each soil property must be separated into at least two components: one for errors due to variability among sampling locations (e.g., pits), and another for errors due to variability among individual samples within locations. When these variance components are known, optimum sampling strategies can be devised by adjusting numbers of sampling sites and numbers of samples per site that will minimize both the variance of the mean and the costs of sample collection.

A similar approach has been used to define sampling strategies for characterizing morphological features. Milfred et al. (1967) evaluated different sampling schemes for estimating the volume percentages of micromorphological features (e.g., clay coatings, nodules, pores, etc.) found in an argillic horizon. Volume percentages were estimated from thin sections made from undisturbed cores. They concluded that, in order to minimize the variance of the feature mean, it was more efficient to increase the number of core samples rather than simply to increase the number of thin sections per core. Murphy and Banfield (1978) found similar results for pore areas, perimeters, and horizontal projections for the B horizons of two soils.

Morphological characteristics that are related to  $K_{sat}$  can be identified using dye tracers. Bouma et al. (1977) used a dye to mark water-conducting voids in clayey soil materials and then related the number and percentages of the dyed voids to the measured  $K_{sat}$  values. One problem with such dye studies is that, if a void contains a neck at some point, then the portion of the void lying downstream of the neck will not be completely filled by the dye, and the total effect of the void on  $K_{sat}$  becomes difficult to determine. Despite this limitation, dye tracers provide useful information about the portion of the material through which water moves.

The objective of this study was to determine the optimum number of sampling pits and number of samples per pit required to estimate mean values for  $K_{sat}$  and water-conducting voids in a 5-ha area of saprolite. Five hectares is a common size for a small suburban subdivision in this area.

### MATERIALS AND METHODS

The study site is in the Piedmont region of North Carolina (Fig. 1a) on the northwest side of Raleigh, in Schenck Forest. The 5-ha area is covered by a dense stand consisting primarily of loblolly pine (*Pinus taeda* L.). The dominant soil series in the area is Cecil (clayey, kaolinitic, thermic Typic Kanhapludult). A mica-schist saprolite occurs within 1.0 m of the soil surface across the site. Based on published soils and geologic maps (Cawthorn, 1970; Parker, 1979) and our

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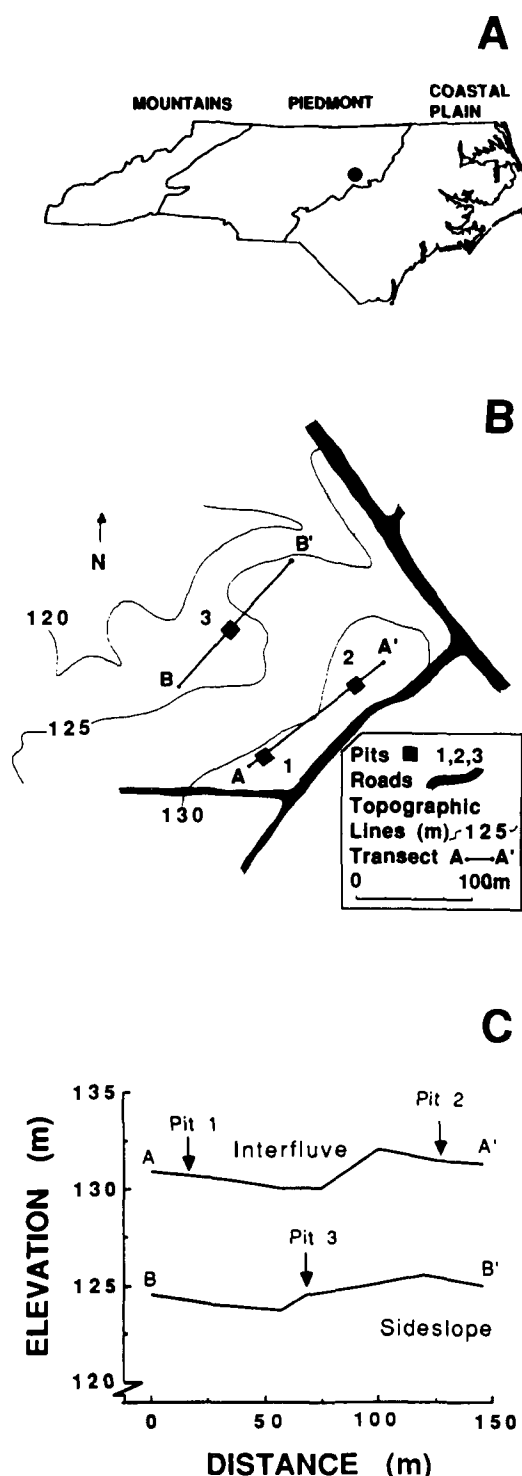


Fig. 1. (A) Location of study site in the North Carolina Piedmont, (B) topographic map of site showing positions of sampling pits, and (C) cross sections of landscape positions where pits were placed. Topographic data from Amoozegar and Hoover (1989).

own preliminary reconnaissance survey, the area appears uniform with respect to soil series and saprolite type.

The study site is in a gently rolling area with about 8 m of relief from the highest to lowest elevations studied (Fig. 1b and c). Approximately 66% of the area consists of an interfluve having a 2 to 6% slope. The remaining area is a linear sideslope, with slopes ranging between 6 and 10%. A

nested experimental design was used for the study (Webster, 1977) with samples collected from three pits and two positions within each pit. Because of constraints imposed by time, money, and landowner preference for minimal soil disturbance, the study had to be limited to three pits. Locations for the sampling pits were selected by landscape position. Pits 1 and 2 were placed on the interfluve (Fig. 1c), and Pit 3 was on the sideslope. Pit locations were chosen to be accessible to a backhoe and to not require removal of large trees. During excavation of Pit 2, a quartz dike was found to bisect the pit. The dike was 0.4 m thick, inclined from the vertical at about 45°, and extended from the soil surface to well below the depth of sampling. Such dikes are known to occur in the saprolite being studied (Parker, 1979) and so the saprolite of Pit 2 was not considered atypical for this deposit.

The pits were dug with a backhoe to be 3 m wide, 5 m long, and 2 m deep to expose both the B horizon and saprolite. Undisturbed cores, oriented vertically, were collected 0.25 m apart in metal rings (76 mm in diam. by 76 mm high) using a hydraulic coring machine (Vepraskas et al., 1990). All cores were carefully examined by eye and those containing apparent cracks or other irregularities caused by sampling were discarded. Twenty-four to 30 cores were collected from the B horizon and a similar number from the saprolite (2-m depth) at each pit. All cores were stored at 4 °C in cardboard containers until analyzed. Samples could not be collected from the dike in Pit 2 because it contained quartz pebbles as large as the ring, which prevented insertion of the ring.

The top and bottom surfaces of each core were carefully picked with a small knife to remove smeared material and to expose natural surfaces and pores. Core bottoms were then covered with cheesecloth and the cores placed in a pan. The level of tap water in the pan was slowly raised during a 48-h period to saturate the samples and minimize air entrapment. When the water level reached the top of the metal ring, the cores were left an additional 24 to 48 h to saturate.

Saturated hydraulic conductivity was determined on 20 cores from the B horizon and 20 cores from the saprolite of each pit using a constant-head permeameter (Hill and King, 1982), again using tap water. The water was partially deaerated by letting it stand in large carboys until it reached room temperature (25 °C). After  $K_{sat}$  was determined, the water-conducting voids in the saprolite samples were marked with a dye. A solution of 0.1% methylene blue dye was allowed to flow through the saturated cores using the constant-head permeameter. Outflow was collected continuously and the concentration of the dye was measured periodically using a spectrophotometer. When the dye concentration in the outflow equaled that of the inflow, the cores were removed from the permeameter and placed in buckets with airtight lids. Time did not permit dyeing of B horizon samples.

Water was removed from the saprolite samples by liquid solvent (acetone) exchange (Murphy, 1986). The saprolite samples were then transferred to metal cans and impregnated under vacuum with an unsaturated polyester resin (Co-Cast, Cope Plastics, Godfrey, IL)<sup>1</sup> diluted with acetone. After the resin had hardened, the samples were randomly divided into two groups of 10 samples each. Horizontal thin sections were then made for one group and vertical sections for the other following standard procedures (Murphy, 1986). Each thin section contained the central portion of the core. The glass slides used (76 by 100 mm) were large enough to contain the entire slice of core for both horizontal and vertical sections.

Four cores from the B horizon and saprolite of each pit

<sup>1</sup> The use of trade names in this publication does not imply endorsement by the North Carolina Agricultural Research Service of the products named, nor criticism of similar ones not mentioned.

were oven dried (105 °C) and bulk density was determined. The soil or saprolite materials were then removed from their rings, composited by position within each pit, and ground to pass a 2-mm sieve. Particle-size distribution was determined on composited samples using the pipette method (Gee and Bauder, 1986) along with dry sieving of sands.

Volume percentages of water-conducting macrovoids and microvoids were estimated for each saprolite thin section by point counting (Chayes, 1956). Macrovoids were channels or natural planes that were much larger (e.g., channel diam.  $\geq 0.1$  mm) than the microvoids (packing voids) found in the groundmass. Water-conducting macrovoids were either the channels and planes whose surfaces were coated by the methylene blue dye or the groundmass adjoining these voids that had been coated with the dye, indicating that flow along the macrovoid had occurred (Fig. 2). Water-conducting microvoids could not be seen as such, and so the volume percentage of the groundmass that was impregnated with dye was determined instead. This *water-conducting groundmass* contained microvoids between mineral particles through which water had flowed. While the amount of water-conducting groundmass was proportional to the volume of water-conducting microvoids, the two features are not the same.

Approximately 250 points were examined for each thin section using a magnification of 160 $\times$  to identify whether the point fell on a dyed macrovoid, dyed groundmass, or other object. This number of points should produce 95% confidence intervals of approximately  $2 \pm 1\%$  and  $50 \pm 6\%$  for properties having estimated volume percentages of 2 and 50%, respectively (van der Plas and Tobi, 1965). All point counts were made by one person using a mechanical stage. Points were spaced uniformly over the entire area of each thin section. In addition, the diameters of channels and widths of planes were measured on 24 thin sections using an ocular scale in the microscope. Diameters were only measured for channels showing a circular cross section, because accurate measurements of channel diameter could be obtained from them.

Analysis of variance procedures (Snedecor and Cochran, 1980) were used to compare mean values for  $K_{sat}$  and volume percentages of water-conducting (dyed) macrovoids and groundmass among the three pits. Frequency distributions of the  $K_{sat}$  data and volume percentages of both water-conducting macrovoids and groundmass showed that these properties were not normally distributed. Therefore, the  $K_{sat}$  values were log-transformed to approximate a normal distribution. An arcsin transformation was used for the point-count results by determining the angle whose sine was equal to the square root of the percentage. The arcsin transformation has been recommended to normalize proportional data (Webster, 1977).

The results from the analysis of variance were used to evaluate how numbers of pits and numbers of cores per pit affected the variance of the sample mean for B horizon  $K_{sat}$  and the  $K_{sat}$  and percentages of water-conducting macrovoids and groundmass of the saprolite. The variance of each mean value ( $V_x$ ) was separated into two components:

$$V_x = S_p^2/n_p + S_c^2/(n_p n_c) \quad [1]$$

where  $S_p^2$  and  $S_c^2$  are the components of the population variance attributable to pits and cores per pit, respectively, and  $n_p$  and  $n_c$  are the numbers of pits and cores per pit, respectively.

An optimum or efficient sampling strategy is determined on the basis of cost as well as  $V_x$  (Snedecor and Cochran, 1980). The total cost ( $C_t$ ) of an investigation can be estimated from

$$C_t = C_p n_p + C_c n_p n_c \quad [2]$$

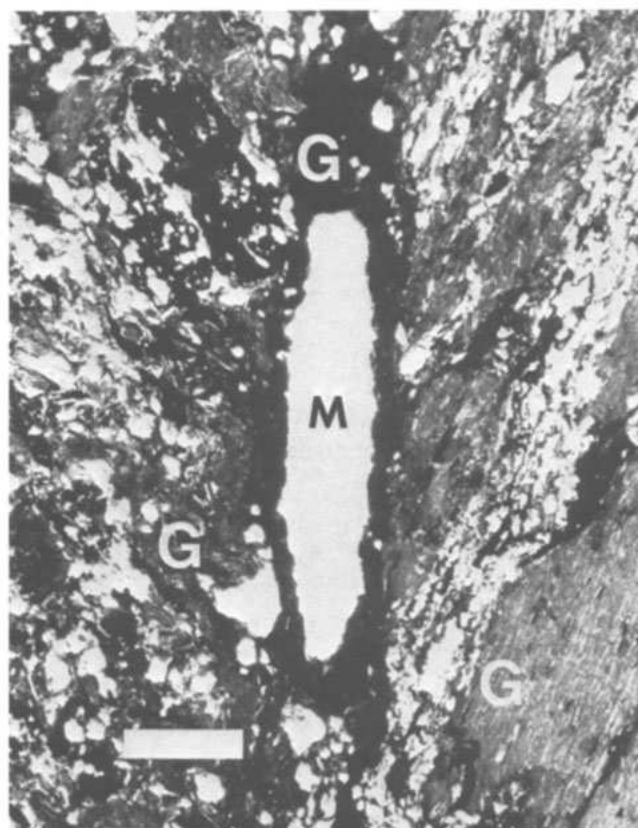


Fig. 2. Example of the water-conducting macrovoids (M) and groundmass (G) that were measured for Pit 1. Scale indicates 0.5 mm, and plane polarized light was used.

where  $C_p$  is the cost of digging and refilling a single pit and  $C_c$  is the cost of collecting a single sample and performing the analysis of interest on it (e.g.,  $K_{sat}$ ).

For this study, the amount of labor (expressed in man-hours) required to perform an operation was of greater interest than the actual cost of it. We assumed that the variables  $C_p$ ,  $C_c$ , and  $C_c$  in Eq. [2] could be replaced by  $T_p$ ,  $T_p$ , and  $T_c$ , respectively, where  $T_t$  is the total amount of time needed to collect and analyze the samples for a given property,  $T_p$  is the time required to dig and refill a single pit, and  $T_c$  is the time required to collect and analyze a single core sample for the property of interest. The optimum sampling strategy (most efficient use of labor) was considered to be the combination of  $n_c$  and  $n_p$  that minimized the product  $V_x T_t$ . Snedecor and Cochran (1980) derived an equation for calculating the  $n_c$  that minimized  $V_x C_t$ . Extending the analogy between costs and time requirements, their equation becomes

$$n_c = \left( \frac{T_p S_c^2}{T_c S_p^2} \right)^{1/2} \quad [3]$$

After the optimum  $n_c$  had been identified, an optimum  $n_p$  for each property was determined on the basis of the size of the confidence interval that was desired to estimate the mean value for each property. The confidence interval was defined as

$$\bar{x} \pm t\alpha/2 \left( \frac{S_p^2}{n_p} + \frac{S_c^2}{n_p n_c} \right)^{1/2} \quad [4]$$

where  $\bar{x}$  is the computed mean for a given property and  $t$  is the percentage points of the Student's  $t$  distribution for the

Table 1. Bulk density, color, and texture for the B horizon and saprolite found in each pit.

| Pit no. | Position  | Bulk density      | Moist color                  | Sand, mm   |            |             |              |              | Silt            | Clay       |
|---------|-----------|-------------------|------------------------------|------------|------------|-------------|--------------|--------------|-----------------|------------|
|         |           |                   |                              | 2.0 to 1.0 | 1.0 to 0.5 | 0.5 to 0.25 | 0.25 to 0.12 | 0.12 to 0.05 | 50 to 2 $\mu$ m | <2 $\mu$ m |
|         |           | Mg/m <sup>3</sup> |                              | %          |            |             |              |              |                 |            |
| 1       | B horizon | 1.27              | 2.5YR 4/6                    | 1.4        | 0.9        | 1.2         | 9.0          | 9.6          | 27.4            | 50.5       |
|         | Saprolite | 1.66              | 2.5YR 8/4                    | 2.2        | 3.1        | 2.4         | 17.0         | 26.3         | 39.5            | 9.5        |
| 2       | B horizon | 1.38              | 2.5YR 4/6                    | 1.7        | 0.9        | 0.7         | 5.6          | 8.4          | 28.7            | 54.0       |
|         | Saprolite | 1.51              | 2.5YR 7/5, and<br>2.5YR 5/8† | 2.7        | 3.3        | 2.7         | 19.4         | 25.6         | 41.7            | 4.6        |
| 3       | B horizon | 1.58              | 2.5YR 5/8                    | 2.4        | 1.3        | 1.4         | 10.8         | 13.4         | 38.9            | 31.8       |
|         | Saprolite | 1.83              | 2.5YR 8/0, and<br>2.5YR 8/6† | 4.5        | 4.2        | 3.0         | 10.5         | 32.4         | 42.7            | 2.7        |

† Dominant color of the saprolite changed across the pit.

$\alpha$  probability level. The limit ( $L$ ) of the confidence interval can be defined as a specific proportion ( $P_x$ ) of the mean by

$$\frac{L}{\bar{x}} = t_{\alpha/2} \left( \frac{S_p^2}{n_p} + \frac{S_c^2}{n_p n_c} \right)^{1/2} / \bar{x} = P_x \quad [5]$$

In this study,  $n_p$  was computed for  $P_x$  values of 0.10, 0.50, and 1.0 for the 95% and 80% confidence intervals.

## RESULTS AND DISCUSSION

Bulk density, color, and particle-size distribution are shown in Table 1 for the B horizon and saprolite of each pit. The textural class of the B horizon is clay for Pits 1 and 2, and clay loam for Pit 3. The textural class of the saprolite is loam for Pit 1 and very fine sandy loam for Pits 2 and 3.

Mean values for  $K_{sat}$  and percentages of water-conducting macrovoids and groundmass are reported and compared among pits in Table 2. Significant differences (0.05 level) among pits were found for all char-

Table 2. Comparison of saturated hydraulic conductivity ( $K_{sat}$ ), water-conducting groundmass, and water-conducting macrovoids among the three pits.

| Pit       | B horizon $K_{sat}$ | Saprolite          |            |            |
|-----------|---------------------|--------------------|------------|------------|
|           |                     | $K_{sat}$          | Groundmass | Macrovoids |
|           |                     | cm h <sup>-1</sup> | %          |            |
| 1         | 0.27a†              | 0.42a              | 40.8a      | 2.4a       |
| 2         | 0.05b               | 0.25ab             | 51.0b      | 0.7b       |
| 3         | 0.18a               | 0.14b              | 52.1b      | 0.6b       |
| $\bar{x}$ | 0.13                | 0.25               | 47.9       | 1.1        |

† In each column, values having the same letter were not significantly different at the 0.05 level as determined on transformed values using Tukey's  $W$  procedure.

Table 3. Variance components, time requirements, and computed optimum number of samples per pit and number of pits needed to estimate confidence intervals (CI) for various properties.

| Property                                         | Variance components           |                        | Time requirements              |                                        | Optimum no. of samples/pit ( $n_c$ ) | Optimum no. of pits/5 ha |     |     |                         |     |     |
|--------------------------------------------------|-------------------------------|------------------------|--------------------------------|----------------------------------------|--------------------------------------|--------------------------|-----|-----|-------------------------|-----|-----|
|                                                  | Among samples/pit ( $S_p^2$ ) | Among pits ( $S_c^2$ ) | Dig and refill a pit ( $T_p$ ) | Collect and analyze a sample ( $T_c$ ) |                                      | $P\bar{x}$ for a 95% CI† |     |     | $P\bar{x}$ for a 80% CI |     |     |
|                                                  |                               |                        |                                |                                        |                                      | 0.1                      | 0.5 | 1.0 | 0.1                     | 0.5 | 1.0 |
| man-hours                                        |                               |                        |                                |                                        |                                      |                          |     |     |                         |     |     |
| B saturated hydraulic conductivity ( $K_{sat}$ ) | 0.43                          | 0.14                   | 2                              | 1.0                                    | 2‡                                   | 156                      | 6   | 2   | 66                      | 3   | 1   |
| Saprolite $K_{sat}$                              | 0.11                          | 0.53                   | 2                              | 1.0                                    | 1                                    | 708                      | 28  | 7   | 299                     | 12  | 3   |
| Saprolite macrovoids                             | 14.4                          | 5.9                    | 2                              | 4.2                                    | 1                                    | 224                      | 9   | 2   | 89                      | 4   | 1   |
| Saprolite groundmass                             | 37.1                          | 11.4                   | 2                              | 4.2                                    | 1                                    | 10                       | 1   | 1   | 4                       | 1   | 1   |

† A specific proportion of the mean ( $P\bar{x}$ ) of 0.1 indicates the size of the confidence limit (one-half of the confidence interval) is equal to 10% of the mean.

‡ Nearest whole integer.

acteristics. Absolute differences in  $K_{sat}$  were generally small. For example, the saprolite  $K_{sat}$  values for Pits 1 and 3 differed by  $<0.3$  cm h<sup>-1</sup>. Differences in amounts of macrovoids between these pits were considered more substantial.

The macrovoids of Pits 1 and 2 were channels and planes that were frequently coated with Fe-Mn oxides and, in a few cases, clay. The oxide coatings frequently plugged the planes. The conductivity of some macrovoids must have been determined by the bottlenecks produced by these coatings. The bottlenecks may also have been responsible for the low  $K_{sat}$  values found in Pit 1 despite its relatively large amount of macrovoids. The channels seen in Pit 3 were generally not coated. The diameters of the channels and widths of open planes measured in thin sections from all pits ranged from 0.1 to 2.2 mm, with most being between 0.2 and 0.8 mm in diameter.

The components of the population variance that were due to variation among cores within a pit ( $S_p^2$ ) and variation among pits ( $S_c^2$ ) are shown in Table 3. Note that, for B horizon  $K_{sat}$ , saprolite macrovoids, and saprolite groundmass, the variance component due to cores within a pit was larger than that due to variation among pits, while for saprolite  $K_{sat}$  the opposite was the case. This indicated that the saprolite  $K_{sat}$  values within a pit were relatively uniform, whereas, for the other properties, a large proportion of the population variance was found within individual pits. This effect can be seen in Fig. 3 where the frequency distributions for the  $K_{sat}$  of the B horizon and saprolite are compared. Within a pit, the range of the  $K_{sat}$  values was narrower for saprolite than the B horizon.

The implications that  $S_p^2$  and  $S_c^2$  have on sampling are shown in Fig. 4, where the  $V\bar{x}$  of saprolite  $K_{sat}$  and percent macrovoids were computed using Eq. [1]. For

saprolite  $K_{sat}$ , the  $V\bar{x}$  was reduced primarily by increasing the numbers of pits. On the other hand, increasing both number of pits and cores per pit markedly reduced  $V\bar{x}$  for percentage of macrovoids. For both properties, as pit number increased the effect that cores per pit had on  $V\bar{x}$  decreased.

The man-hour requirements needed to dig and refill pits and also to collect and analyze samples are shown in Table 3. In order to minimize  $V\bar{x} T_i$ , taking only one core per pit was most efficient for all saprolite properties, while two cores per pit were best for B horizon  $K_{sat}$ . This was expected for saprolite  $K_{sat}$  because increasing  $n_c$  had relatively little effect on the variance of the saprolite  $K_{sat}$  mean (Fig. 4). While increasing  $n_c$  produced greater reductions in  $V\bar{x}$  for macrovoids than for  $K_{sat}$  (Fig. 4), collecting and analyzing a sample required more than twice as much time as digging a pit. This was due in large part to the time-consuming nature of making thin sections. Thus, the most efficient way to minimize  $V\bar{x} T_i$  for all saprolite properties was to increase the number of pits and to decrease the number of samples analyzed.

The optimum or most efficient number of pits varied with the property of interest, the desired size ( $P\bar{x}$ ) of the confidence interval, and the desired level of confidence in the estimate (Table 3). Water-conducting groundmass of saprolite was the least variable property and required relatively few pits to estimate a 95% confidence interval for the different  $P\bar{x}$  values. Estimating 95% confidence intervals having  $P\bar{x}$  values of 0.1 for the other properties required sampling extremely large numbers of pits. The pit requirements could be greatly reduced by accepting wider confidence limits ( $P\bar{x} = 0.5$  or 0.1) or less reliable ones (80% confidence interval). Clearly, the desired size and ac-

curacy of the confidence limit will vary with the needs of the investigation.

The results reported in Table 3 reflect not only the natural variability of the properties in the landscape, but also the experimental techniques employed. For example, the sample size used to measure the property affects the variance of the final measurements (Bear, 1972). The ideal sample size (volume) for each property is termed the representative elementary volume (REV). Measurements made on samples having a volume  $< REV$  have a greater variance than measurements made on samples whose volume is  $\geq REV$  (Bear, 1972). When measurements are made on samples having a size  $< REV$ , the number of pits needed to estimate a property for a preset level of precision is also higher due to the greater variance. Based on the results shown in Table 3, the volume of the core samples used may have been  $\geq REV$  for water-conducting groundmass, but  $< REV$  for the remaining properties.

Digging large pits to extract a single core sample at depths of approximately 2 m is impractical. A hydraulic coring machine may be used instead to extract undisturbed cores in metal rings at the desired depth. The coring machine disrupts less soil or saprolite than a backhoe, and its use would probably be more acceptable to landowners. Pits are useful when morphological data such as profile descriptions or other analyses are required. For studies which require that numerous locations be sampled or that detailed profile

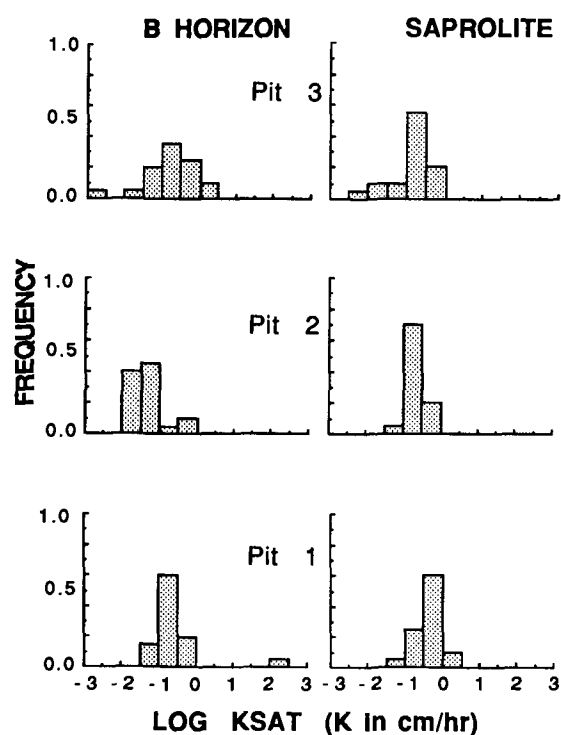


Fig. 3. Frequency distributions for saturated hydraulic conductivity ( $K_{sat}$ ) in the B horizon and saprolite of each pit.

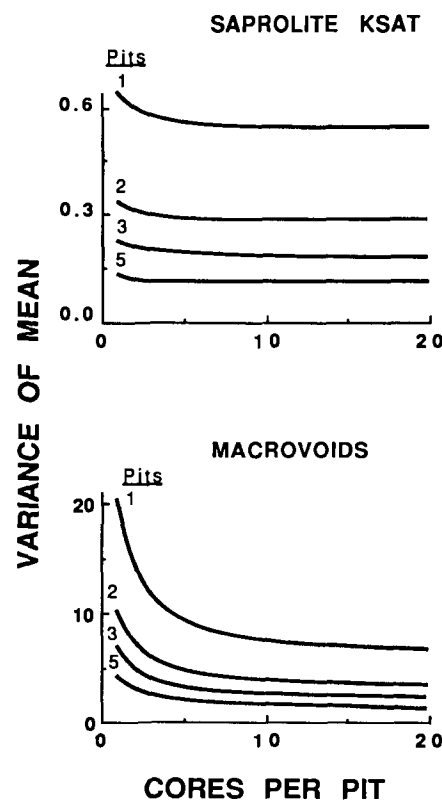


Fig. 4. Relationship of variance of sample mean to number of sampling pits and number of cores per pit for saprolite saturated hydraulic conductivity ( $K_{sat}$ ) and percentage of macrovoids. Variances were computed for transformed values. Note that a curve for five pits is included for illustration.

descriptions be obtained, then a combination of both pits and hydraulically extracted cores may be the best approach to use to collect samples. The locations of the sampling pits or extracted cores should be carefully selected to represent an area. A stratified sampling design (Webster, 1977) can be used to ensure that samples are collected from representative topographic positions and saprolite deposits.

For example, stratified random sampling could be used to measure the  $K_{\text{sat}}$  and volume percentages of water-conducting voids of another mica-schist saprolite across one or two landscape positions. To do this, we would estimate mean values using 95% confidence intervals having a  $P\bar{x}$  of approximately 0.5. This would require that at least 28 core samples be collected to estimate a  $K_{\text{sat}}$  mean, with fewer samples required for the other properties (Table 3). To collect samples from a 5-ha area, the site would first be divided (for convenience) into 25 square cells measuring 45 m on a side. Two core samples would be collected randomly from each cell using a hydraulic probe to extract samples. The  $K_{\text{sat}}$  would be measured on all 50 samples. Water-conducting voids would be measured on a smaller number of samples (e.g., 15) selected by pooling five adjacent cells with similar saprolite, and randomly drawing three core samples from the 10 collected from each group of the pooled cells. Equations for estimating the mean and variance for this stratified sampling procedure are given by Snedecor and Cochran (1980) and Webster (1977).

In summary, estimating mean values for the  $K_{\text{sat}}$  and volume percentages of water-conducting voids in this saprolite was done most efficiently by collecting only one sample per sampling location (two samples for B horizon  $K_{\text{sat}}$ ) and to let the number of sampling locations be determined by the desired size and precision of the confidence interval. While a nested experimental design was used for the study, results showed that a stratified sampling design would be more efficient with sampling locations spread over the area of interest. Because we included two different landscape positions in our sampling scheme, the variance due to pits was probably higher than it would have been if only one landscape position had been sampled. Sampling-location requirements can probably be reduced

without affecting the size and precision of the confidence intervals by restricting sampling to only one landscape position.

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